

de novo (upper dot and bar)

guided (lower dot and bar)

same family rule in both modes; Rustle-internal, not a tool comparison

a Pairwise (gene pairs)

b One-to-one bipartite (family members)

c Reference families, per mode

Human A119b (T2T-CHM13 v2.0)

held out, never used

chr6 · Compara, primates
15 families · 35 genes
chr6 · Soto 2025
3 families · 6 genes

held out, reused verdict set

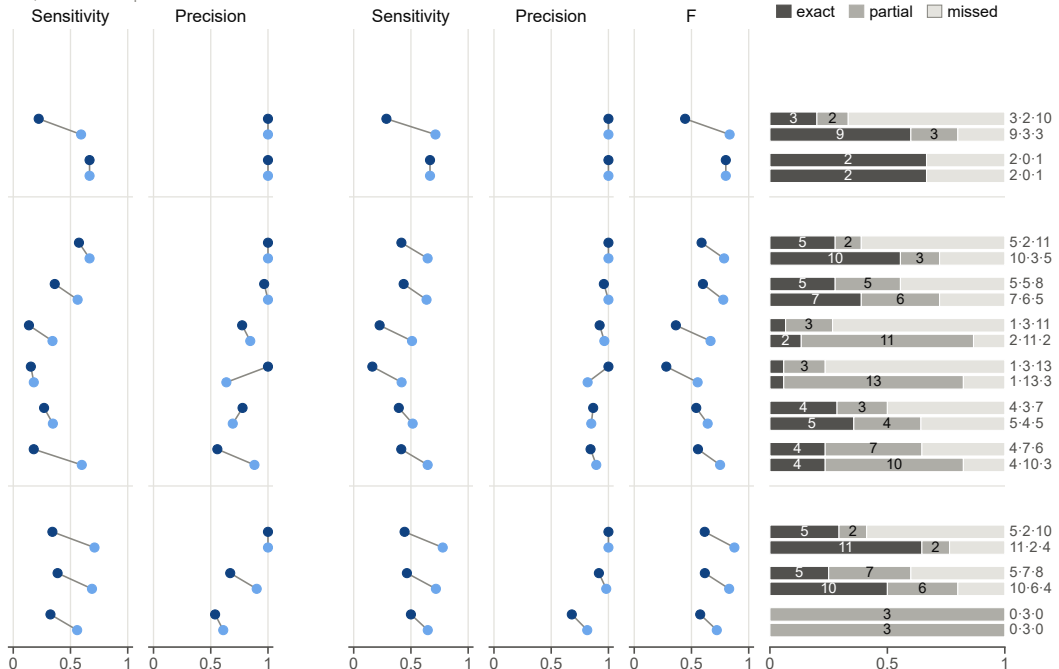
chr2 · Compara, primates
18 families · 48 genes
chr2 · Soto 2025
18 families · 55 genes

chr8 · Compara, primates
15 families · 53 genes
chr8 · Soto 2025
17 families · 43 genes

chr10 · Compara, primates
14 families · 33 genes
chr10 · Soto 2025
17 families · 65 genes

development

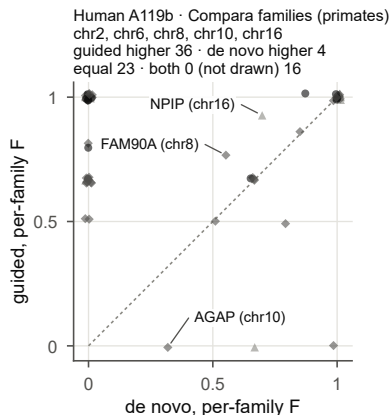
chr16 · Compara, primates
17 families · 54 genes
chr16 · Soto 2025
20 families · 71 genes
chr16 · NPIP reference set
3 families · 34 genes



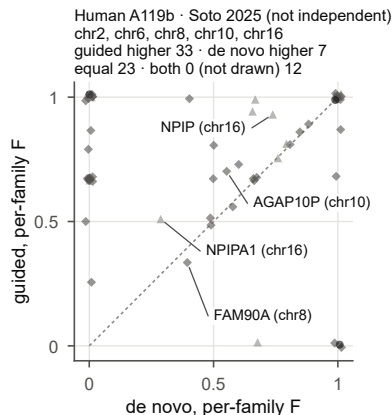
Per-family F: one point per reference family that at least one mode recovers (jitter ≤ 0.015)

● held out, never used ◆ held out, reused verdict set ▲ development

d



e



f

